# Project Title: CampusSwap - Academic Resource Exchange System

## Team Members


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## Overview

CampusSwap is a web-based system designed for university students to exchange academic resources. After logging in, users can create, update, and delete posts to share resources. To find materials, users can browse and search posts, and filter them by department. Students can also send exchange requests to other users and manage these requests by accepting or rejecting them. The main goal of this system is to make sharing educational resources easy, organized, and accessible for all students.

## Entities and Attributes

**User:** Members of the platform have a unique user_id, username, email, password, and an avatar name. Each user is also associated with a specific department through a dept_id. The user_id is the key attribute. A user can have multiple skills at the same time (Multivalued).

**Resource:** This entity represents the materials being shared. Each resource has a unique resource_id, a type (e.g., PDF, book), and a description. It is also categorized by a dept_id. The resource_id is the key attribute.

**Post:** Users create posts to describe what they are offering and what they are requesting in return. A post includes a unique post_id, the user_id of the owner, an offer_id, a request_id, and the current status of the listing. The post_id is the key attribute.

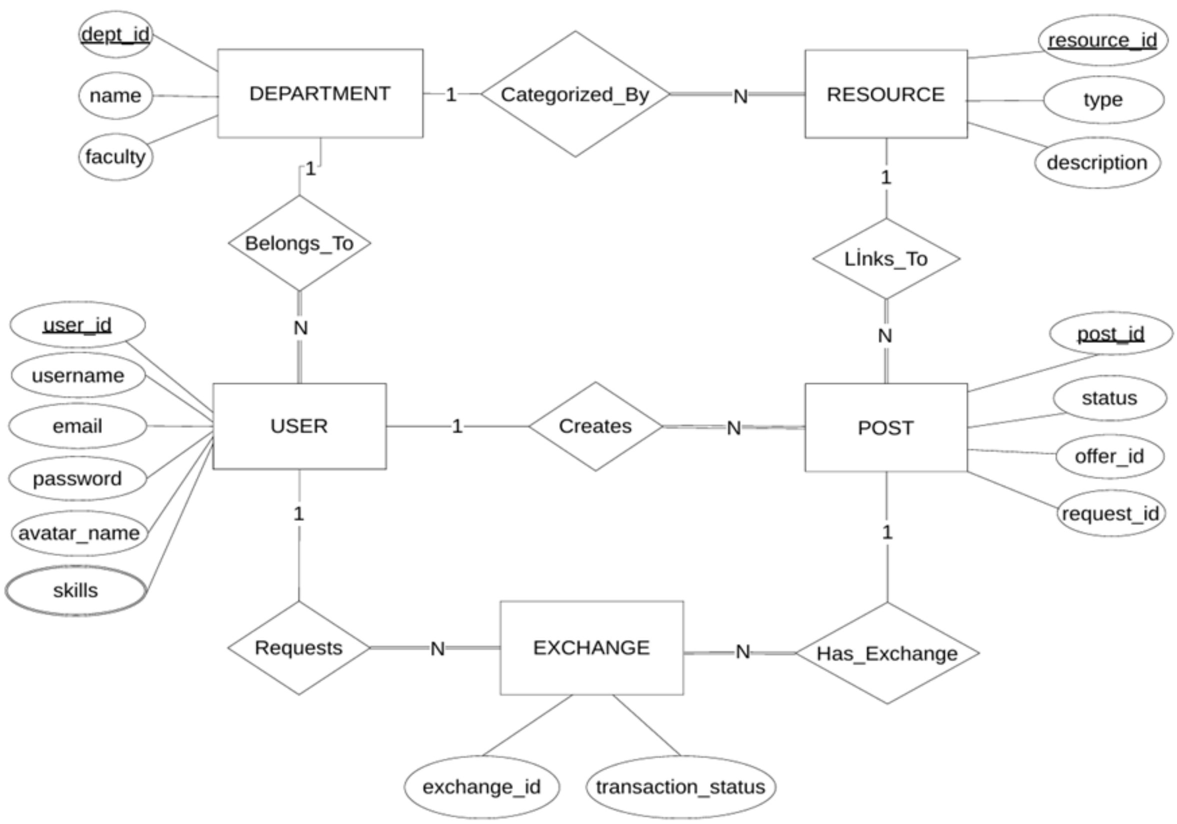
**Exchange:** This entity manages the trade requests. It tracks a unique exchange_id, the related post_id, the requester_id (the user asking for the trade), and the transaction status. The exchange_id is the key attribute.

**Department:** To organize users and resources, the system stores department information. Each department has a unique dept_id, a name, and belongs to a specific faculty. The dept_id is the key attribute.

**Relationships:**

- A user can create multiple posts; but each post belongs to exactly one owner (1:N).
- A post links to specific resources; many posts can use the same resource from the system (1:N).
- Multiple users can request a trade for a single post; a post can have multiple exchange requests (1:N).
- A department can have many users; one user can have one department (1:N).
- A department can have many resources; a resource can have one department (1:N).

## ER DIAGRAM



# Normalization

For the normalization process, an initial, unnormalized structure was created first. This structure represents a large, mixed report containing user profiles, their skills, department details, and their posts together. Then, it was broken down step by step up to 3NF.

## Initial Unnormalized Form (UNF)

A single, massive table was drafted where repeating groups of user skills are kept inside a general user and post record.

UNF_SYSTEM_REPORT (User_ID, Username, Email, Password, Avatar_Name, Dept_ID, Dept_Name, Faculty, {Skill}, Post_ID, Offer_ID, Request_ID, Status)

## Functional Dependencies (FDs)

Before applying the normal forms, the functional dependencies within the system were identified:

- User_ID ->Username, Email, Password, Avatar_Name, Dept_ID
- Dept_ID->Dept_Name, Faculty
- Resource_ID ->Type, Description, Dept_ID
- Post_ID ->User_ID, Offer_ID, Request_ID, Status
- Exchange_ID ->Post_ID, Requester_ID, Transaction_Status

**Step 1: First Normal Form (1NF)**

- **What was fixed:** Repeating groups were eliminated.
- **Why:**Since each attribute must hold an atomic value, the repeating "skills" attribute was split into a new table. This new table is linked back using a composite primary key.
- **Resulting Tables:**
  - SYSTEM_1NF (User_ID, Username, Email, Password, Avatar_Name, Dept_ID, Dept_Name, Faculty, Post_ID, Offer_ID, Request_ID, Status)
  - USER_SKILLS_1NF (User_ID, Skill)

**Step 2: Second Normal Form (2NF)**

- **What was fixed:** Partial dependencies and logically distinct groups were separated.
- **Why:** In the 1NF table, User details only depend on User_ID, and Post details only depend on Post_ID. To establish proper primary keys and eliminate partial dependencies, we extract Posts from the general system report.
- **Resulting Tables:**

  - USER_2NF (User_ID, Username, Email, Password, Avatar_Name, Dept_ID, Dept_Name, Faculty)
  - USER_SKILLS_2NF (User_ID, Skill)
  - POST_2NF (Post_ID, User_ID, Offer_ID, Request_ID, Status)

**Step 3: Third Normal Form (3NF) & Final Schema**

- **What was fixed:** Transitive dependencies were resolved.
- **Why:** Non-key attributes shouldn't depend on other non-key attributes. Since Dept_Name and Faculty are found via Dept_ID, they were separated completely into a Department table. The remaining entities (Resources, Exchanges) are established with the same logic. The final tables now perfectly match the ERD structure.

## Final Normalized Tables (3NF)

- USERS (PK: user_id, username, email, password, avatar_name, FK: dept_id)
- USER_SKILLS (PK/FK: user_id, PK: skill)
- DEPARTMENTS (PK: dept_id, name, faculty)
- RESOURCES (PK: resource_id, type, description, FK: dept_id)
- POSTS(PK: post_id, FK: user_id, FK: offer_id, FK: request_id, status)
- EXCHANGES (PK: exchange_id, FK: post_id, FK: requester_id, transaction_status)